$L_1 = L_2 = L = 0,3345\ \mu H/m$

$L_m = 0,0372\ \mu H/m$

$C_1 = C_2 = C = 110,83\ pF/m$

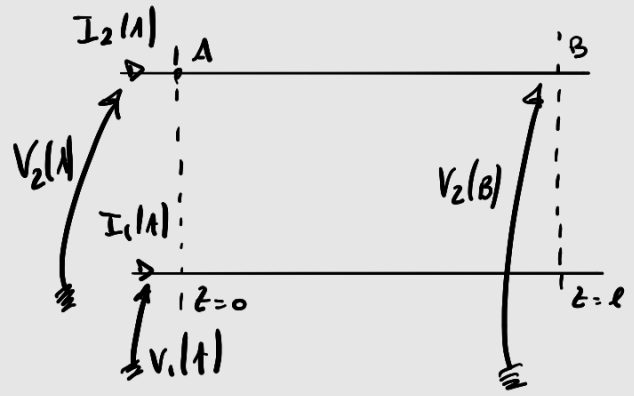
$C_m = 4,93\ pF/m$

$\ell = 5\ m$

$I_1(1) = 4\ mA \qquad I_2(1) = 0$

$V_1(1) = 0,8V \qquad V_2(1) = 0,2V$

$f_0 = 7,5\ GHz \Rightarrow \omega = 2\pi f_0 \approx 47 \cdot 10^9\ rad/s$



$V_2(B) = ?$

Since the line is lossless and symmetrical, then we can divide the system in even and odd mode:

$C_e = C = 110,83\ pF/m$

$L_e = L + L_m = 0,3717\ \mu H/m$

$C_o = C + 2C_m = 120,69\ pF/m$

$L_o = L - L_m = 0,2973\ \mu H/m$

We calculate now the characteristic impedances and the phase constants for both modes at the given frequency.

$$Z_e = \sqrt{\frac{L_e}{C_e}} = \sqrt{\frac{L+L_m}{C}} = \sqrt{\frac{0,3717 \cdot 10^{-6}}{110,83 \cdot 10^{-12}}} = 57,912\ \Omega$$

$$Z_o = \sqrt{\frac{L_o}{C_o}} = \sqrt{\frac{L-L_m}{C+2C_m}} = \sqrt{\frac{0,2973 \cdot 10^{-6}}{120,69 \cdot 10^{-12}}} = 53,806\ \Omega$$

$$k_e = \omega\sqrt{C_e L_e} = \omega\sqrt{C(L+L_m)} = 302,459\ rad/m$$

$$k_o = \omega\sqrt{C_o L_o} = \omega\sqrt{(C+2C_m)(L-L_m)} = 282,276\ rad/m$$

Since we have a 2 conductors, lossless and symmetrical line, we can write the transformation matrix $\underline{\underline{M}}$ as:

$$\underline{\underline{M}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

With this we can find modal voltages by using the following relationship:

$V^m(0) = M^{-1} V(0)$

$I^m(0) = M^{-1} I(0)$

With this we can write that:

$V^e(A) = \frac{1}{\sqrt{2}}\left(V_1(A) + V_2(A)\right) = \frac{1}{\sqrt{2}}(0,8 + 0,2) = 0,707\ V$

$V^o(A) = \frac{1}{\sqrt{2}}\left(V_1(A) - V_2(A)\right) = \frac{1}{\sqrt{2}}(0,8 - 0,2) = 0,424\ V$

$I^e(A) = \frac{1}{\sqrt{2}}\left(I_1(A) + I_2(A)\right) = \frac{1}{\sqrt{2}}\left((4 \cdot 10^{-3}) + 0\right) = 0,00283\ A$

$I^o(A) = \frac{1}{\sqrt{2}}\left(I_1(A) + I_2(A)\right) = I^e(A) = 0,00283\ A$

We propagate voltages at the end of the line by using Telegrapher's equations. We evaluate them both for the even and odd line and then we calculate the final voltage.

$V_0^{e+} = \frac{1}{2}\left[V^e(A) + Z_e I^e(A)\right] = \frac{1}{2}\left[0,707 + (57,312 \cdot 0,00283)\right] = 0,435\ V$

$V_0^{e-} = \frac{1}{2}\left[V^e(A) - Z_e I^e(A)\right] = \frac{1}{2}\left[0,707 - (57,312 \cdot 0,00283)\right] = 0,272\ V$

$V^e(B) = V_0^{e+} e^{-jk_e l} + V_0^{e-} e^{jk_e l} = 0,435\angle(-302,453 \cdot 0,05) + 0,272\angle(302,453 \cdot 0,05) =$

$= -0,591 - 0,0834j\ V$

$V_0^{o+} = \frac{1}{2}\left[V^o(A) + Z_o I^o(A)\right] = \frac{1}{2}\left[0,424 + (53,806 \cdot 0,00283)\right] = 0,288\ V$

$V_0^{o-} = \frac{1}{2}\left[V^o(A) - Z_o I^o(A)\right] = \frac{1}{2}\left[0,424 - (53,806 \cdot 0,00283)\right] = 0,136\ V$

$V^o(B) = V_0^{o+} e^{-jk_o l} + V_0^{o+} e^{jk_o l} = 0,288\angle(-282,276 \cdot 0,05) + 0,136\angle(282,276 \cdot 0,05) =$

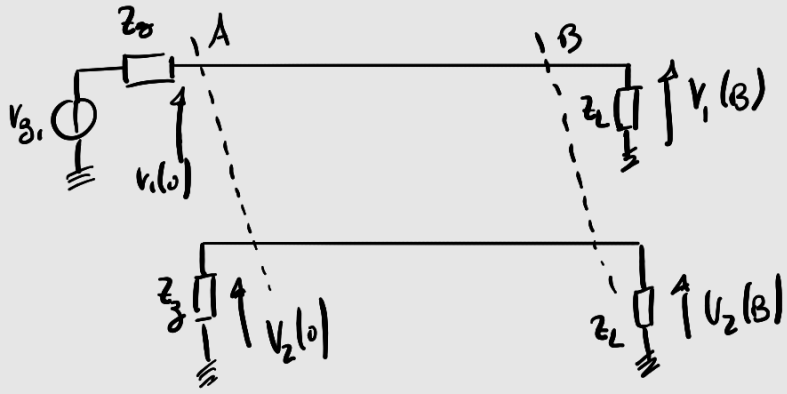
$= 0,00991 - 0,152j\ V$

Now we use the relation: $V(B) = M V^m(B)$ to find $V_2(B)$

$V_2(B) = \frac{1}{\sqrt{2}}\left(V^e(B) - V^o(B)\right) \approx -0,425 + 0,0449j\ V$

$|V_2(B)| = 0,428\ V$

Exercise 2.



$L_1 = L_2 = L = 0,4484\ \mu H/m$

$L_m = 0,1543\ \mu H/m$

$C_1 = C_2 = C = 65,26\ pF/m$

$C_m = 22,92\ pF/m$

$\frac{V_2(0)}{V_1(0)} = ?$

$AB = 12\ cm$

$f = 6\ GHz \Rightarrow \omega = 2\pi f = 37,69 \cdot 10^9\ rad/s$

$Z_g = Z_L = Z_0 \rightarrow$ isolated line

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{0,4484 \cdot 10^{-6}}{65,26 \cdot 10^{-12}}} = 82,89\ \Omega$$

We decompose the microstrip in even and odd line modes.

$C_e = C = 65,26\ pF/m$

$L_e = L + L_m = 0,6027\ \mu H/m$

$C_o = C + 2C_m = 111,1\ pF/m$

$L_o = L - L_m = 0,2941\ \mu H/m$

$$Z_e = \sqrt{\frac{L_e}{C_e}} = 96,101\ \Omega$$

$k_e = \omega\sqrt{L_e C_e} = 236,431\ rad/m$

$$Z_o = \sqrt{\frac{L_o}{C_o}} = 51,451\ \Omega$$

$k_o = \omega\sqrt{L_o C_o} = 215,494\ rad/m$

We calculate reflection coefficients in B

$$\Gamma_B^e = \frac{Z_0 - Z_e}{Z_0 + Z_e} = \frac{82,89 - 96,101}{82,89 + 96,101} = -0,0738$$

$$\Gamma_B^o = \frac{Z_\infty - Z_o}{Z_\infty + Z_o} = \frac{82,89 - 51,451}{82,89 + 51,451} = 0,234$$

knowing the lenght of the line, we can move the reflection coefficient from the B to the A section.

$$\Gamma_A^e = \Gamma_B^e e^{-2jk_e l} = -0,0738\angle(-2 \cdot 236,431 \cdot 0,12) = -0,0724 + 0,0143j$$

$$\Gamma_A^o = \Gamma_B^o e^{-2jk_o l} = 0,234\angle(-2 \cdot 215,494 \cdot 0,12) = 0,0275 - 0,232j$$

With this and the characteristic impedence of each mode line we can define the input impedences:

$$Z_{in}^e = Z_e \frac{1+\Gamma_A^e}{1-\Gamma_A^e} = 83,093 + 2,389j\ \Omega$$

$$Z_{in}^o = Z_o \frac{1+\Gamma_A^o}{1-\Gamma_A^o} = 48,663 - 23,883j\ \Omega$$

Since we need $\frac{V_2(0)}{V_1(0)}$, we don't care of the value of voltage V_g is generating, because it will cancel out eventually. For the further calculations we'll use a normalized generator of 1V.

To know how the voltage is divided on the conductors we have to use the voltage divider formula, knowing that (with the 1V generator) model sources are both equal to $\frac{1}{\sqrt{2}} = V_g^e = V_g^o$

$$V_A^e = V_g^e \frac{Z_{in}^e}{Z_{in}^e + Z_\infty} = \frac{1}{\sqrt{2}} \frac{83,093 + 2,389j}{83,093 + 2,389j + 82,89} = 0,354 + 0,00508j\ V$$

$$V_A^o = V_g^o \frac{Z_{in}^o}{Z_{in}^o + Z_\infty} = \frac{1}{\sqrt{2}} \frac{48,663 - 23,883j}{48,663 - 23,883j + 82,89} = 0,276 - 0,0783j\ V$$

$$V_1(A) = V_1(0) = \frac{1}{\sqrt{2}}(V_A^e + V_A^o) = 0,445 - 0,0518j\ V$$

$$V_2(A) = V_2(0) = \frac{1}{\sqrt{2}}(V_A^e - V_A^o) = 0,0552 + 0,0589j\ V$$

$$\left|\frac{V_2(0)}{V_1(0)}\right| = 0,18$$